

A Data-driven approach for investigating crime in the Canton neighborhood of Michigan

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Currently, the Canton Police Department in Michigan has 30 years of big data, including calls and reported crime. In the age of Artificial Intelligence (AI), this data can serve as a cornerstone for analyzing criminal activity in this neighborhood. It can give us a better idea of what has happened and what will happen regarding calls and crimes. It can help us manage our resources, such as police allocation and surveillance cameras, more productively. It can guide us toward better objectives for police officers, ensuring they make the town safer with the least amount of crime. Based on these motivations, here, we propose a data-driven approach to investigate the crime in the Canton neighborhood of Michigan.

We have organized this proposal as follows. We discuss the business understanding of it in Section 1. In Section 2, we describe how we will understand the data. We outline a brief plan for preparing the dataset for predictive modeling in Section 3. In Section 4, we propose the types of testing, techniques, or models we expect to conduct. We discuss the plan for deployment of the model in Section 5. Outline of our plan is shown in Fig1.

1 Business Understanding

Canton was a small town. However, populations have grown over the last 30 years. The crime rates have also increased with the increasing population. We need more police as well as resources like police allocation, surveillance cameras, etc to make the town safer with the least amount of crime. It can be very expensive if we do not manage and apply our resources efficiently. Our current data-driven approach can significantly reduce this cost by helping the Police Department make better decisions and be more efficient.

2 Data Understanding

The backbone of every data-driven approach is the quality of data we have. The Canton Police Department provides a rich dataset. This dataset has three different types of information: (1) Calls, (2) Dispositions, and (3) Census.

2.1 Calls

In Calls, all calls for the past ten years are available. For example, if someone files a complaint for an assault, that would be one record. This data has been collected from three different databases. The data has been incorporated into one database, but it does have inconsistencies within its categories. This data is available in the form of four columns:



Figure 1: Schematic of the Data-driven approach for investigating crime in the Canton neighborhood of Michigan

- Subzone (common between Calls and Census Data)
- Call_number (common between Calls and Dispositions).
- Complaint
- Date_Received

2.2 Dispositions

Dispositions deal with the results of the call/complaint. For example, if someone files a complaint for an assault, then the disposition would be what happened in the end, such as a report filed or an arrest. Dispositions data has two columns:

- Call_Number (common between Calls and Dispositions)
- DISPOSITION

2.3 Census

Census has been recorded for each subzone. The Canton town is divided into these subzones. Census data has 35 columns:

- | | |
|-------------------------------------|-----------------------------------|
| • Subzone | • Total time to work (minutes) |
| • SqFootage | • Drove alone to work |
| • HousingUnits | • Carpooled to work |
| • Population | • Population_3andover |
| • Population_Male | • Enrolled in school (nursery) |
| • Population_Female | • Preschool |
| • Worked | • Enrolled in kindergarten |
| • Worked in a principal city | • Enrolled in college |
| • Worked outside any principal city | • Undergraduate years |
| • Workers who travel to work | • Graduate or professional school |

- Not enrolled in school
- Households_earnings
- Households_wageorsalaryincome
- Households_selfemploymentincome
- Households_interest_dividends
- Households_SSI
- Households_publicassistanceincome
- MedianAge_Total
- MedianAge_Male
- MedianAge_Female
- HouseholdIncome_Median
- HouseholdIncome_Median_under25
- HouseholdIncome_Median_25to44
- HouseholdIncome_Median_45to64
- HouseholdIncome_Median_65andover
- Income_PerCapita

In addition to these available data, we can pull data from external sources.

3 Data Preparation

As we described in the previous section, we have a rich, large amount of data. However, these data may not be in a model-ready form. We need to process the data. First, we read the data from the file using a pandas dataframe(<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.html>). After we successfully read the data, we will investigate the dataframe using the ".info()" command. This will tell us how many rows and columns are present in the dataframe. We will try to detect missing values and duplicate values and remove them. We will visualize important features by their distribution. We will remove the outliers if we find them. We will convert the categorical values to floats or integers by encoding them before passing them to the model.

4 Modeling

After understanding and processing the data, we will develop the machine learning model, which can be either a regression or a classification model <https://www.ibm.com/think/topics/classification-vs-regression>, depending on the predicted feature. If the computed quantity is continuous, we will develop a regression model. On the other hand, if it is categorical, we will construct a classification model.

We will extensively validate the model before its deployment for the production run. We will split the dataset into a training dataset and a test dataset. We evaluate the model on our test dataset.

5 Deployment

We will deploy the model after validating it. We expect our model to answer many important questions, so that the Police Department can make better decisions and be more productive. For example, which crimes were registered as most frequent, and from which subzone? Once our developed model learn these statistics from the past, it should predict the probability of these crimes in the future.